Positivity of the Fibonacci characteristic function

A solution to MathDB 360139

Principia Math In Progress

18 September 2026

Abstract

The characteristic generating function of the letter a in the Fibonacci fixed word is strictly positive on $(-1, 1)$. We prove this by telescoping periodic-prefix approximants and bounding their negative increments using the spacing of Fibonacci numbers. An elementary instance of the source's even-block decomposition handles the remaining compact interval. This proves Conjecture 5.12 of Pouti, Ramsey, and Strungaru.

Task 1: the original problem and its full solution

Let $\sigma(a) = ab$, $\sigma(b) = a$, and let $w = w_0 w_1 \cdots = abaababa \cdots$ be its fixed word starting in a. The characteristic function in [1, Sections 3–5] is

$$C_a(z) = \sum_{j \geq 0} \mathbf{1}_{\{w_j = a\}} z^j, \qquad |z| < 1.$$

In particular the constant coefficient is 1. All series below are considered inside their disk of absolute convergence.

Theorem 1. *For every real $x \in (-1, 1)$, one has $C_a(x) > 0$. Moreover, for $9/10 \leq t < 1$,*

$$C_a(-t) > \frac{1}{1+t} - \frac{127(1+t)}{600t^2} \geq \frac{2753}{48600(1+t)} > \frac{2753}{97200}.$$

An exact telescoping expansion

Write $A_{-1} = b$, $A_0 = a$, and $A_{n+1} = A_n A_{n-1}$ for $n \geq 0$. Then $A_n = \sigma^n(a)$ for $n \geq 0$. Put $L_n = |A_n| = F_{n+2}$, where $F_0 = 0, F_1 = 1$, and set

$$P_n(z) = \sum_{\substack{0 \leq j < L_n \\ (A_n)_j = a}} z^j, \qquad f_n(z) = \frac{P_n(z)}{1 - z^{L_n}} \quad (n \geq 0).$$

Here $P_{-1} = 0$, $P_0 = 1$, and

$$P_{n+1} = P_n + z^{L_n} P_{n-1}. \tag{1}$$

Since the A_n are increasing prefixes of w, $P_n(z) \to C_a(z)$ and $f_n(z) \to C_a(z)$ for $|z| < 1$.

Define

$$D_n = P_n(1 - z^{L_{n-1}}) - P_{n-1}(1 - z^{L_n}), \qquad n \geq 0.$$

Equation (1) gives $D_0 = 1 - z$ and $D_{n+1} = -z^{L_n} D_n$. Since $\sum_{j=0}^{n-1} L_j = L_{n+1} - 2$, induction proves

$$D_n = (-1)^n z^{L_{n+1}-2}(1 - z). \tag{2}$$

Let $a = L_n$, $b = L_{n-1}$, $c = 2a + b = L_{n+2}$, and $d = 3a + 2b = L_{n+3} = 2c - a$. The word identity $A_{n+3} = A_n A_{n-1} A_n A_n A_{n-1}$ gives

$$P_{n+3} = P_n(1 + z^{a+b} + z^{2a+b}) + P_{n-1}(z^a + z^{3a+b}).$$

Subtracting after a common denominator and using (2) yields

$$\begin{aligned} P_{n+3}(1 - z^a) - P_n(1 - z^d) &= -z^a(1 + z^c)D_n \\ &= (-1)^{n+1} z^{c-2}(1 + z^c)(1 - z). \end{aligned} \tag{3}$$

For $n = 3k$, all of a, b, c, d are odd, because F_m is even exactly when 3 divides m. Thus, for $0 < t < 1$,

$$f_{3k+3}(-t) - f_{3k}(-t) = (-1)^k E_k(t),$$

where

$$E_k(t) = \frac{(1 + t)t^{c_k-2}(1 - t^{c_k})}{(1 + t^{a_k})(1 + t^{2c_k-a_k})} > 0, \qquad a_k = F_{3k+2}, \quad c_k = F_{3k+4}. \tag{4}$$

The bound $E_k(t) \le 2t^{c_k-2}$ makes the series of increments absolutely convergent for each fixed $t < 1$: the strictly increasing integers c_k satisfy $c_k \ge k + 3$. Taking limits in the telescoping sum therefore gives the exact identity

$$C_a(-t) = \frac{1}{1 + t} + \sum_{k\ge0} (-1)^k E_k(t). \tag{5}$$

A bound for widely spaced negative increments

Lemma 1. *Let $x_0, x_1, \ldots > 0$ satisfy $x_{\ell+1} \ge 16x_\ell$, and put $g(u) = u(1 - u)/(1 + u)^2$. Then*

$$\sum_{\ell\ge0} g(e^{-x_\ell}) < \frac{127}{600}.$$

Proof. For $0 < u < 1$, the inequality $g(u) \le 1/8$ follows from

$$\frac{1}{8} - g(u) = \frac{(1 - 3u)^2}{8(1 + u)^2} \ge 0.$$

For $x > 0$, also

$$g(e^{-x}) = \frac{e^{-x}}{1 + e^{-x}} \tanh(x/2) \le \frac{x}{4}, \qquad g(e^{-x}) \le e^{-x}.$$

Here $\tanh y \le y$ for $y \ge 0$, as follows by differentiating. Split the terms into $x < 1/4$, $1/4 \le x < 4$, and $x \ge 4$. The first group is finite, and summing backwards from its largest element gives

$$\sum_{x_\ell<1/4} g(e^{-x_\ell}) < \frac{1}{4} \frac{1/4}{1 - 1/16} = \frac{1}{15}.$$

The same strict bound holds if this group is empty. The middle group contains at most one element, hence contributes at most $1/8$. In the last group, the successive arguments are at least $4 \cdot 16^r \geq 4 + 60r$, so its contribution is at most

$$\sum_{r \geq 0} e^{-4-60r} = \frac{e^{-4}}{1 - e^{-60}} < \frac{1}{50}.$$

Indeed, $e^4 > \sum_{j=0}^{7} 4^j/j! = 16319/315 > 51$ and $e^{60} > 51$, which imply the last strict inequality. Adding the three bounds gives $1/15 + 1/8 + 1/50 = 127/600$. □

By the arithmetic–geometric mean inequality,

$$(1 + t^a)(1 + t^{2c-a}) \geq (1 + t^c)^2.$$

Consequently (4) implies

$$E_k(t) \leq \frac{1+t}{t^2} g(t^{c_k}). \tag{6}$$

The negative increments in (5) have $c_{2\ell+1} = F_{6\ell+7}$. For $m \geq 7$,

$$F_{m+6} = 13F_m + 8F_{m-1} > 17F_m > 16F_m,$$

since $F_{m-1} > F_m/2$. Thus $x_\ell = -F_{6\ell+7} \log t$ meets Lemma 1. Discarding the positive increments in (5) gives

$$C_a(-t) > \frac{1}{1+t} - \frac{127(1+t)}{600t^2}. \tag{7}$$

For $9/10 \leq t < 1$, $(1+t)/t \leq 19/9$ and

$$1 - \frac{127}{600}\left(\frac{19}{9}\right)^2 = \frac{2753}{48600} > 0.$$

This proves all the quantitative assertions of Theorem 1.

The remaining compact interval

We now reprove the needed low-order case of the even-block decomposition in [1, Section 5]; that decomposition is an established input, not a new claim. There is no occurrence of bb in w: every substitution image starts in a, and a b only occurs at the end of $\sigma(a)$. Since $w = \sigma^3(w)$, group the letters of the latter occurrence of w into successive pairs, starting at its beginning, and apply σ^3. With $A = \sigma^3(a) = abaab$ and $B = \sigma^3(b) = aba$, this expresses w as a concatenation of the blocks AB, AA, BA. Each block has even length. Their a-characteristic polynomials evaluated at $-t$ are, respectively,

$$\begin{aligned} R(t) &= 1 + t^2 - t^3 - t^5 - t^7, \\ S(t) &= (1 + t^2 - t^3)(1 - t^5), \\ T(t) &= (1 + t^2)(1 - t^3) + t^6. \end{aligned}$$

The latter two are positive for $0 \leq t < 1$. Also

$$R'(t) = t(2 - 3t - 5t^3 - 7t^5).$$

The parenthesis is strictly decreasing on $[0,\infty)$; thus R first increases and then decreases, and its minimum on $[0,9/10]$ occurs at an endpoint. Both endpoint values are positive:

$$R(0)=1, \qquad R(9/10)=\frac{122131}{10000000}.$$

All offsets of the concatenated blocks are even. Hence their contributions to $C_a(-t)$ are nonnegative powers of t times positive numbers for $0 \le t \le 9/10$. The first block already contributes a positive number. Absolute convergence permits this grouping, proving $C_a(-t) > 0$ on this interval.

Finally, $C_a(x) \ge 1$ for $0 \le x < 1$, because its coefficients are nonnegative and its constant term is 1. Together with (7), this proves Theorem 1 on the entire required interval.

Tasks 2–5: proposed follow-up directions

These are questions for further work, not additional solved claims.

Task 2. Determine the set of limit points of $C_a(-t)$ as $t \to 1^-$, including exact or rigorously characterized values of its lower and upper limits.

Task 3. Determine the optimal constant $c_* = \inf_{0\le t<1} C_a(-t)$ and whether that infimum is attained at an interior point.

Task 4. For each integer $m \ge 2$, let $w^{(m)}$ be the fixed word starting in a of $a \mapsto a^m b$, $b \mapsto a$. Classify the m for which its a-characteristic generating function is positive throughout $(-1,1)$.

Task 5. Determine the largest $\rho \in (0,1]$ for which $C_a(z)$ has no zero in $|z| < \rho$, and, if $\rho < 1$, rigorously locate and count all zeros with modulus ρ.

Verification and status. The accompanying verifier checks the polynomial identities against directly substituted words, the exact constants and block polynomials, and finite telescoping sums in rational arithmetic. These checks supplement the analytic proof for the infinite series. The current 23-page source was read in full; its authors list the paper as accepted or in press in 2026. The dated literature audit records the bounded prior-claim search. Verification is by the authoring agent only, without independent peer review.

References

[1] A. Pouti, C. Ramsey, and N. Strungaru, *Generating functions of substitutions*, arXiv:2309.16100v2, 25 May 2026, 23 pp.; accepted/in press, Rocky Mountain Journal of Mathematics, according to the authors' publication lists checked 18 September 2026. The target is Conjecture 5.12 on p. 21.